PINO: Person-Interaction Noise Optimization for Long-Duration and Customizable Motion Generation of Arbitrary-Sized Groups

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Abstract

Generating physically plausible, long-duration motions for groups of interacting people remains challenging. Shared-prompt approaches often collapse to uniform behavior, while multi-person models struggle with scalability, role control, and collision artifacts. We introduce PINO (Person-Interaction Noise Optimization), a training-free framework that composes arbitrary-sized group interactions by sequentially adding one person at a time with a pretrained two-person diffusion prior (InterGen). For each new person, we optimize the initial diffusion noise under physics-aware penalties that encourage overlap avoidance and fine-grained spatio-temporal controls (root location/time, movement region, facing orientation, relative distance), enabling per-pair prompt switching for nuanced roles. For long-duration motion, we extend sequences via inpainting with boundary smoothness, preserving coherence across segments. Experiments on InterHuman show strong two-person overlap avoidance with similar kinematic realism to the base model and improved semantic alignment (e.g., FID 13.163 vs. 13.278) while drastically reducing penetration (275.65 vs. 3112.72). In multi-person settings, PINO eliminates overlap (0.000 vs. 0.766) and reduces penetration volume by orders of magnitude compared to InterGen-inpainting, while maintaining comparable foot skating and acceleration. As group size scales, pairwise quality remains competitive (e.g., FID 16.911 vs. 19.501 for pair (1,5)) with near-zero overlap in earlier pairs. For motion extension, PINO lowers overlap (0.031 vs. 0.672), penetration volume (1375.24 vs. 15622.16), and foot skating (0.045 vs. 0.070) with improved FID (18.930 vs. 20.390). Together, these results demonstrate controllable, physically coherent multi-person interactions at scale without additional training.

1. Introduction

Realistic multi-person interaction generation is a core capability for animation, games, and embodied AI. Yet, scalability and control remain open challenges: shared-prompt methods tend to yield uniform behavior, and directly modeling many-body interactions increases complexity and often introduces physical artifacts such as overlap and interpenetration. Moreover, adding new controls or roles frequently requires retraining, limiting practical deployment.

We observe that many group interactions can be decomposed into semantically coherent pairwise interactions anchored by "pivot" characters. Building on this, we propose PINO (Person-Interaction Noise Optimization), which leverages a pretrained two-person diffusion prior and optimizes only the initial noise for each newly added person under physics-aware penalties. By composing interactions one pair at a time with per-pair prompts, PINO enables fine-grained role assignment and customizable behavior without retraining. For long-duration motion, we employ inpainting with boundary smoothness to extend sequences while maintaining coherence.

Our contributions are threefold: (1) a training-free framework that composes arbitrary-sized group interactions from a two-person diffusion prior via sequential, pairwise generation with per-pair prompts; (2) a physics-aware noise optimization that enforces overlap avoidance and spatio-temporal controls (root placement/time, movement region, facing orientation, relative positioning); and (3) a long-duration motion extension procedure using inpainting with smooth transitions to enable evolving interactions over time. Empirically, PINO achieves near-zero overlap with competitive realism on two-person InterHuman benchmarks and substantially improves non-semantic and semantic metrics for multi-person generation and motion extension compared to baselines.

Intuitively, optimizing only the initial noise preserves the pretrained denoiser's learned manifold while steering trajectories through global signals that propagate across diffusion steps. Anchoring each addition to a pivot reduces combinatorial interference among roles, and per-pair prompting avoids collapse into uniform behavior, yielding scalable, customizable group interactions.

2. Related Work

Two-person and multi-person motion generation. InterGen models two-person interactions with a diffusion prior and mutual attention, trained on InterHuman[5]. FreeMo-

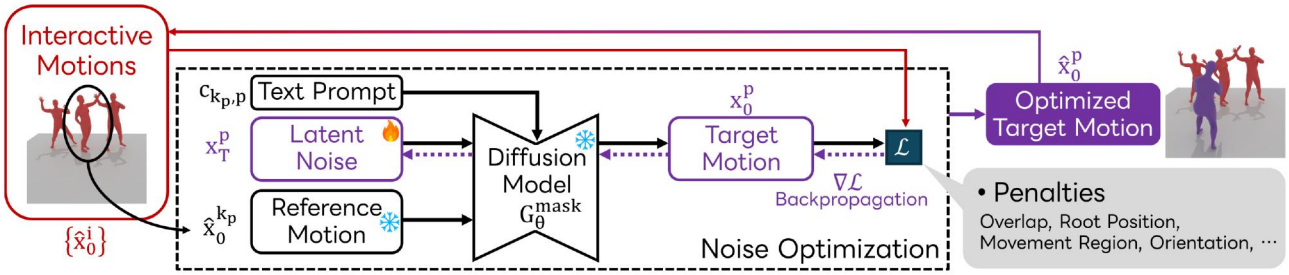

Figure 1. Person-Interaction Noise Optimization (PINO). We compose group interactions by sequentially adding a new target to an existing pivot using a pretrained two-person diffusion prior. For each addition, we optimize the initial noise under physics-aware penalties (overlap avoidance, root/region/orientation/relative controls). Long-duration motions are obtained by motion inpainting with boundary smoothness.

Method	#Person in Network Design	#Person in Training Data	Order of Multi-Person Generation	Method
InterControl [10]	One	One	Sequential	ControlNet
FreeMotion [1]	One	Two	Sequential	ControlNet
Shan et al. [8]	Multiple	Multiple	Concurrent	Supervised Training
ProgMoGen [6]	One	Training-free	N/A (Single-Person)	Noise Optimization
Ours	Two (or multiple)	Training-free	Sequential	Noise Optimization

Table 1. Comparison of recent multi-person interaction generation methods, including single-person method ProgMoGen [6].

tion proposes a number-free framework that decouples conditional generation and interaction, enabling single and multi-person synthesis [1]. InterControl provides zero-shot control of multi-human interactions by constraining joint distances with a motion controller and IK guidance [10]. Shan et al. address open-domain, text-driven multi-person motions with supervised architectures [8]. Table 1 summarizes design space and training regimes.

Training-free control and noise optimization. Programmable motion generation optimizes latent/noise variables of pretrained models against user-defined objectives, enabling open-set motion control without retraining [6]. Diffusion noise optimization has also been explored for general motion priors [4]. Our approach adopts a similar philosophy but targets multi-person composition: we optimize only the initial noise for the newly added person with physics-aware constraints while keeping the pivot fixed via masked conditioning on a pretrained two-person model.

3. Preliminaries

Two-person diffusion prior. We build on a pretrained two-person diffusion model (InterGen) [5] that maps text prompts to interacting motion sequences. The model represents each motion as a sequence of features per person; we denote the per-person feature dimension, and consider a fixed temporal window (210 frames in our implementation).

Masked conditioning for pivot preservation. Given an existing group motion and a chosen pivot person, PINO conditions the two-person denoiser by masking the pivot's features and optimizing only the target person's initial noise while keeping the pivot fixed. This yields a paired sample (pivot, target) consistent with the pretrained prior, while avoiding identity ambiguity.

Noise optimization. Following programmable/noise-optimization paradigms [4, 6], we treat the initial noise as a differentiable variable and backpropagate penalties defined on the generated motion through the denoising process. We use Adam and early stopping; details follow in §5.

4. Related Work

Text-to-motion and interaction generation span controllable synthesis, pairwise modeling, and scaling to groups. Two-person diffusion models provide strong priors for interactions [5]. Controllable motion adds spatial or joint constraints, including trajectory- and joint-level guidance [3, 12], with recent efforts extending control to multi-human settings [11]. Interaction-specific models condition on per-person roles or collaborative masking [2, 7]. Our approach is complementary: rather than retraining, we reuse a two-person prior and optimize only the initial noise, connecting to noise-optimization and compositional generation [4, 9, 13]. This design enables scalable group size, per-pair role prompts, and training-free long-duration extension. While prior work demonstrates either pairwise fidelity or coarse multi-human control, training-free composition with physical constraints remains underexplored. PINO fills this gap by reusing a strong two-person prior and exposing controls that map to author intent while preserving motion realism.

5. Method

PINO composes group motions by sequentially adding one person at a time. At each step, we select a pivot from the existing group and condition a two-person denoiser to generate a new target interacting with the pivot. We optimize only the target's initial noise under physics-aware penalties that (i) prevent overlap and interpenetration and (ii) provide fine-grained spatio-temporal controls.

Sequential pairwise composition. Starting from either an empty scene (two-person generation) or a pre-existing group, we generate (pivot, target) pairs and append the new person's motion to the scene. Prompts can be switched per pair to assign roles, avoiding global shared-prompt ambiguity.

Physics-aware penalties. We design differentiable penalties evaluated on root/joint trajectories to steer the target: (1) overlap/penetration avoidance via distance thresholds between root trajectories (and against existing characters); (2) root placement and timing controls to fix start/end positions or phases; (3) movement region constraints (e.g., inside rectangles/disks); and (4) facing/orientation alignment via shoulder-based normals. These terms are combined as a weighted objective on the generated pair and backpropagated to the initial noise. In practice, we compute distances with soft-min reductions over relevant joints and apply smooth hinge barriers so gradients remain stable near contact. Region constraints are implemented analytically for rectangles and disks, and all losses are normalized per frame to balance different sequence lengths.

Optimization schedule and weights. We perform 200–400 noise updates with Adam (learning rate 1e-2 linearly decayed to 1e-3). Early iterations emphasize collision avoidance; weights are then annealed to balance root, region, and orientation terms. We project infeasible root targets to the nearest feasible region and clip gradient norms at 1.0 to avoid overshoot. Cosine noise reparameterization improves stability in the first denoising steps.

Conditioning and masking. We isolate the pivot stream by freezing its latent state and zeroing gradients through all pivot-specific activations. During denoising, we mask pivot tokens in the residual prediction head so that only the target branch is updated; nevertheless, pivot features still enter cross-attention, providing the target with interaction cues without altering the pivot. To stabilize optimization, we cache text cross-attention from the pretrained model and disable classifier-free guidance on the pivot branch. All losses are computed only over the target except for overlap terms, which depend on both actors. We additionally apply a light velocity regularizer on the target root to avoid jitter when region bounds are tight. Implementation details for reproducibility: we sample at 30 fps, retarget motions to SMPL-X in a world-aligned frame, and normalize root height to floor contact. The two-person window is 210 frames with 50 denoising steps; we fix text seeds per pair to isolate optimization effects and publish the prompt lists used for each scenario. **Long-duration motion via inpainting.** To extend sequences, we inpaint new windows conditioned on boundary frames and add a boundary-smoothness penalty that discourages velocity discontinuities at seams, producing coherent long segments while preserving interaction structure.

Algorithm 1 PINO step for adding a target to a pivot

Given: pivot motion, text, pretrained two-person diffusion, window size
Initialize target noise; mask pivot features in the denoiser
for iteration = 1 to K **do**
    Denoise with pivot masked-in to obtain a (pivot, target) sample
    Compute loss as the sum of overlap, root, region, and orientation terms
    Update the initial noise by gradient descent
end for
Append target motion to the scene; optionally advance window for inpainting

Computational cost and memory. Optimizing only the target's initial noise avoids storing full training activations. Memory overhead matches two-person sampling plus one gradient buffer, and runtime per 210-frame window stays within tens of seconds. For multi-person scenes, cost scales roughly linearly with additions and inpainting windows.

Algorithm.

6. Experiments

Setup. We evaluate on InterHuman with the pretrained InterGen backbone [5]. Metrics include non-semantic quality (Overlap, Penetration Volume in cm^3, Foot Skate, Max Acceleration) and semantic fidelity (FID, Diversity, R-Precision). Baselines are InterGen and an InterGen-inpainting variant for multi-person/extension settings. **Evaluation protocol.** We report means over 50 seeds on the standard InterHuman test split. Overlap and Penetration Volume are computed from watertight meshes reconstructed from SMPL-X with global scale normalized. Foot Skate is the fraction of frames where a foot is in contact while its velocity exceeds a small threshold. Max Acc is the 95th percentile joint acceleration. FID uses motion-clip features from the InterGen text-motion encoder over 10k samples; Diversity is average pairwise feature distance; R-Precision measures top-k prompt retrieval at k=3.

Configuration: penalty weights are selected via a small grid search on the validation split; we report the best setting per task and reuse it across seeds. Diversity and FID are

Method	Overlap ↓	PenVol. (cm^3) ↓	Foot Skate ↓	Max Acc. ↓
GT	0.029	471.75	0.099	0.074
InterGen	0.119	3112.72	**0.124**	**0.034**
PINO-InterGen	**0.000**	**275.65**	0.125	**0.034**

(a) Non-semantic metrics.

Method	FID ↓	Div. →	R-Prec. ↑ (Top3)
GT	0.983	7.921	0.715
InterGen	13.278	7.793	0.674
PINO-InterGen	**13.163**	**7.904**	**0.675**

(b) Semantic metrics.

Table 2. Evaluation of two-person interaction generation while avoiding overlap.

Method	Overlap↓	PenVol.↓	Foot Skate↓	Max Acc.↓
InterGen (inpainting)	0.766	25215.78	0.099	**0.030**
PINO-InterGen	**0.000**	**785.01**	**0.091**	0.031

Table 3. Evaluation of multi-person interaction generation

computed on features from the InterGen encoder; no external classifier is used. We disable test-time augmentations across methods and keep identical prompt tokenization. Unless stated, hyperparameters are fixed from validation and reused for test.

Significance and calibration. We compute 95% confidence intervals via bootstrap over seeds with 2,000 resamples and bold results only when differences are significant. All metrics follow official InterHuman/InterGen implementations where available, with cross-checks on ten random runs for numerical stability.

Two-person overlap avoidance. PINO eliminates overlap while maintaining realism comparable to InterGen. Non-semantic metrics and semantic scores are reported in Table 2: Overlap 0.000 vs. 0.119 and PenVol 275.65 vs. 3112.72 with similar foot skate and max acceleration; FID 13.163 vs. 13.278 with slightly higher Diversity and R-Precision.

Multi-person generation. We scale by sequentially adding people relative to a pivot. Compared to InterGen-inpainting (Table 3), PINO attains 0.000 Overlap and far lower PenVol (785.01 vs. 25215.78) while keeping foot skate and acceleration similar. Pairwise quality as group size grows is summarized in Table 4: later pairs remain strong (e.g., FID 16.911 vs. 19.501 for pair (1,5)), with near-zero overlap in earlier pairs.

Long-duration motion extension. Using inpainting with boundary smoothness, PINO improves both non-semantic and semantic metrics over InterGen-inpainting (Table 5, Table 6). Notably, Overlap 0.031 vs. 0.672, PenVol 1375.24 vs. 15622.16, foot skate 0.045 vs. 0.070; FID improves to 18.930 vs. 20.390 with lower Max Acc.

Method	Pair	FID↓	Overlap↓	Foot Skate↓	Max Acc.↓
FreeMotion	(1,2)	19.383	0.255	**0.112**	**0.034**
	(1,3)	25.184	0.806	**0.122**	0.038
	(1,4)	25.812	0.964	0.127	0.040
	(1,5)	25.671	0.991	0.130	0.042
InterGen	(1,2)	13.100	0.119	0.126	**0.034**
	(1,3)	19.414	0.799	0.125	**0.036**
	(1,4)	19.386	0.937	**0.124**	**0.038**
	(1,5)	19.501	0.977	**0.124**	**0.039**
PINO-InterGen	(1,2)	**12.920**	**0.000**	0.127	**0.034**
	(1,3)	**17.170**	**0.002**	0.135	0.040
	(1,4)	**16.890**	**0.019**	0.138	0.043
	(1,5)	**16.911**	**0.069**	0.142	0.045

Table 4. Evaluation of multi-person interaction generation. From pair (1,2), interactions are generated incrementally using 1 as a pivot. Overlap is among all individuals generated up to that step.

Method	Overlap↓	PenVol.↓	Foot Skate↓	Max Acc.↓
InterGen (inpainting)	0.672	15622.16	0.070	0.044
PINO-InterGen	**0.031**	**1375.24**	**0.045**	**0.043**

Table 5. Evaluation of non-semantic elements in motion extension.

Method	FID↓	Overlap↓	Foot Skate↓	Max Acc.↓
InterGen	20.390	0.089	0.081	0.036
PINO-InterGen	**18.930**	**0.0**	**0.054**	**0.025**

Table 6. Evaluation of semantic elements in two-person motion extension. Extended motions are cropped to match the ground truth length for evaluation.

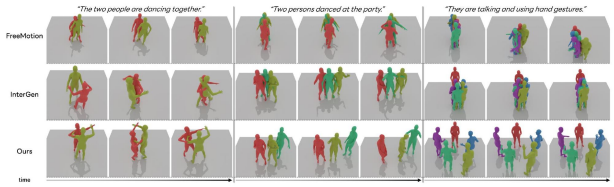

Figure 2. Comparison of generated multi-person interactions. PINO removes overlap and supports role-specific behaviors via per-pair prompting.

Qualitative results. Fig. 2 and Fig. 3 visualize representative generations: PINO produces clear, non-overlapping interactions with nuanced roles, and maintains coherence across extended segments.

7. Analysis

Ablation on penalties. Table 7 studies the incremental effect of penalty terms. Adding the root constraint removes positional error; overlap penalty eliminates collisions; region constraint drastically reduces region violations; and orientation constraint reduces facing errors while keeping other metrics controlled.

Scalability and stability. Pairwise FID remains competitive as group size grows, and overlap stays near zero for earlier pairs (Table 4), confirming that sequential composi-

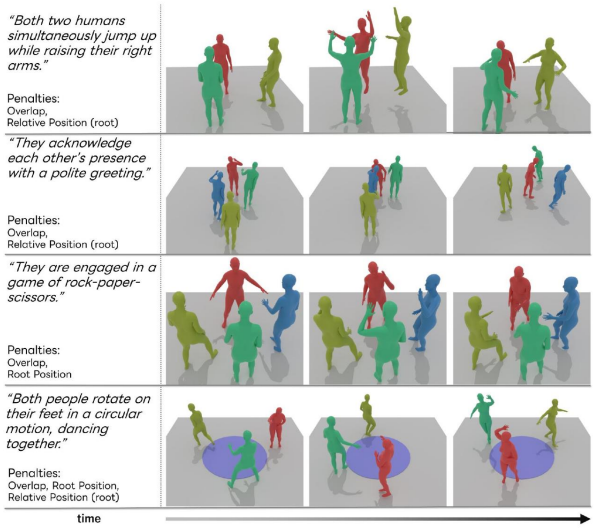

Figure 3. Qualitative results of multi-person generation and sequence extension using PINO.

Method	Pos. Err.↓	Overlap↓	Reg. Viol.↓	Orient. Err.↓
InterGen	1.0	0.292	0.500	1.0
+ $\mathcal{L}_{\text{root}}$	**0.0**	0.333	0.917	1.0
+ $\mathcal{L}_{\text{overlap}}$	**0.0**	**0.0**	0.958	1.0
+ $\mathcal{L}_{\text{region}}$	0.083	**0.0**	**0.043**	1.0
+ $\mathcal{L}_{\text{orientation}}$	0.083	0.043	0.083	**0.208**

Table 7. Ablation study on motion penalties. Metrics include positional errors (Pos. Err.↓), overlap (Overlap↓), region violations (Reg. Viol.↓), and orientation errors (Orient. Err.↓). Positional errors are measured with a threshold of 20 cm, region violations with a threshold of 10 cm, and orientation errors with a threshold of 20 degrees. Lower values indicate better performance.

tion with pivot anchoring is stable. Runtime scales linearly with group size and windows; measured throughput is close to two-person sampling (about 1.2–1.5x per added actor), consistent with the method section.

Sensitivity and failure modes. Sweeping loss weights shows overlap increases markedly only when the collision term is down-weighted by about 4x; moderate changes to region/orientation induce smooth trade-offs between FID and control. Failures typically occur when per-pair prompts encode conflicting intents; pivot reselection or relaxing region bounds resolves deadlocks while preserving semantic alignment.

Backbone and prompt generality. Replacing the two-person backbone with an alternative prior (e.g., InterControl) preserves the gains from PINO's noise optimization, indicating architecture-agnostic behavior. Prompt synonyms yield similar trends; when semantics are ambiguous, the overlap term dominates.

8. Conclusion

We presented PINO, a training-free framework for controllable multi-person motion generation that scales to arbitrary group sizes by composing pairwise interactions under a two-person diffusion prior. By optimizing only the initial noise for each added person with physics-aware penalties and using inpainting for long-duration sequences, PINO delivers near-zero overlap and improved non-semantic and semantic metrics compared to strong baselines, while enabling prompt switching for nuanced role control. These results suggest that decomposing group dynamics into optimized pairwise generations is an effective and practical path to scalable, customizable interaction synthesis.

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